

DUAL-Compass: Direction-Aware Simultaneous Learning and Unlearning for Explainable VLM Hallucination Mitigation

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Abstract

Hallucination in vision–language models (VLMs) is usually treated as an unsigned rate to be minimized, so mitigation applies a single anti-over-claim pressure uniformly. We argue instead that it is a *signed* phenomenon: a model pool mixes over-claiming (*yes-bias*) and over-withholding (*no-bias*), which require *opposite* parameter updates. We present **DUAL-Compass**, which turns a direction-aware diagnosis of a model’s own errors into the training signal: the errors are partitioned into an *unlearn* stream (over-claims, suppressed with negative-preference optimization), a *learn* stream (missed objects, reinforced with supervised updates), and a *retain* stream, optimized *simultaneously*, while a compass-guided closed loop re-diagnoses the signed score during training and reduces the dominant error direction toward balance (mitigation, not exact zeroing on held-out data). Across three VLMs (2B to 7B), the loop improves held-out adversarial accuracy with zero collapses over nine runs, whereas single-direction pressure, fixed mixing, balanced fine-tuning, and an EFUF-style baseline each fail in a distinct way. A controlled stress test shows that both negative-preference and gradient-ascent unlearning collapse under fixed weights yet stay stable under the loop, so the failure lies in missing feedback rather than the loss. The diagnosed direction is distribution-dependent: on the official POPE benchmark (COCO), re-diagnosing a model on that distribution repairs it (the 2B model rises from 73.0% to 91.3%), and a blind human audit confirms both the diagnosis (90% agreement) and the repair (85% preferred over inference-time and EFUF-style baselines). Code, data, and trained adapters are available (anonymized for review) at <https://anonymous.4open.science/r/DUAL-Compass-DD12>.

CCS Concepts

• Computing methodologies → Artificial intelligence; • Information systems → Data mining.

Keywords

vision–language models, hallucination, machine unlearning, trustworthy machine learning, explainable diagnosis, preference optimization

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1 Introduction

Large vision–language models (VLMs) hallucinate: they describe objects that are not in the image, and they overlook objects that are [11, 13, 16, 24]. As VLMs move into deployment (assistive description for blind users, content moderation, robotics, medical triage), both failure directions carry real cost, though of different kinds: asserting what is absent makes a system *unsafe* (false alarms, fabricated evidence), while denying what is present makes it *unhelpful* (missed objects, incomplete descriptions). A rapidly growing literature measures hallucination [4, 8, 19, 25, 27, 28, 33] and mitigates it [9, 12, 14, 15, 34–38, 40]. Almost all of this work shares one structural assumption: hallucination is a *magnitude*, a rate to be pushed down. The corrective pressure is therefore one-directional, typically suppressing the model’s tendency to over-claim.

Recent diagnostic work challenges this assumption. HalluCompass [2] showed that aggregate metrics such as POPE’s yes-ratio arithmetically cancel two qualitatively different failure modes, *yes-bias* (over-claiming absent objects) and *no-bias* (withholding supported evidence), and that prompt-level interventions chosen against the diagnosed direction are strictly harmful: applying a chain-of-thought prompt to a yes-biased model raised its false-positive rate by up to 23.3 percentage points. Related observations of sycophantic assent bias in language models [26] reinforce the picture that assertion bias is a directional, model-specific property. However, that line of work stops at *evaluation*: the diagnosis selects a prompt, not a parameter update. Prompt-mode routing is superficial in the sense that the model’s weights, and thus its underlying propensity to hallucinate, are unchanged; the intervention must be re-applied at every inference call and can be silently dropped by downstream users.

On the training side, machine unlearning, originally developed for privacy-motivated data removal [5, 6, 31], has recently been applied to hallucination mitigation. EFUF [30] uses CLIP similarity to locate hallucinated spans in captions and applies gradient ascent to unlearn them; follow-up work moves the unlearning target into the vision encoder to combat superficial forgetting [3]. Yet these methods inherit the unsigned view in three ways. First, the signal that selects *what* to unlearn is an opaque scalar (a CLIP similarity threshold): it cannot state what *kind* of failure is being removed. Second, only the over-claiming direction is treated; nothing in the objective *teaches* the model the objects it currently misses. Third, the mixing between forgetting and retention is fixed by hand. General-purpose unlearning research meanwhile shows that forgetting and retaining need not conflict; they can be optimized simultaneously

with aligned gradients [1, 7, 32, 39], though in that literature the forget set is externally given (privacy, copyright), not produced by a diagnosis of the model’s own errors.

This paper closes the gap between the two lines. We propose **DUAL-Compass** (Direction-aware Unlearning-And-Learning), a training framework in which an explainable, signed hallucination diagnosis becomes the supervision itself. Our contributions are:

- **C1, Diagnosis as supervision.** The model’s own errors on a presence-probing pool are split by direction into an *unlearn* stream (over-claims), a *learn* stream (missed objects), and a *retain* stream (correct behavior). Because every training example carries its diagnostic direction, the intervention is auditable at the level of individual failure types: one can state which direction of error was suppressed, which was reinforced, and why the balance shifted during training (Section 4.1).
- **C2, A simultaneous, direction-symmetric objective.** A single three-term loss applies *opposite gradient pressures at the same time*: negative-preference optimization (NPO) [39] suppresses hallucinated assertions, supervised learning reinforces missed detections, and a retain term anchors utility (Section 4.2). Our ablations show this simultaneity is a *requirement*, not a convenience: each single-direction pressure in isolation is either ineffective or actively harmful (Section 5.2).
- **C3, Compass-guided closed-loop control.** Fixed mixing weights overshoot the balance point, converting a yes-biased model into a no-biased one. DUAL-Compass treats the signed score as a control variable: the model is re-diagnosed every K steps, the stream-sampling probabilities are re-set from the current score, and training terminates when the probe-measured imbalance enters a dead-band, mitigating the dominant direction rather than necessarily zeroing it on held-out data (Section 4.3).
- **C4, Direction-balanced data expansion.** We mine per-image bias-target annotations into new presence probes, doubling supervision to 8,006 queries while growing *both* directional streams, a property no preference-pair corpus provides (Section 4.4).

Figure 1 summarizes the framework.

Findings. With LoRA adapters [10] on Qwen2-VL-2B, Qwen2.5-VL-3B [29], and LLaVA-1.5-7B [18], the experimental picture is consistent across models. (i) The best fixed-weight DUAL run reduces the held-out over-claim rate from 80.9% to 5.5% while raising accuracy from 64.0% to 86.0%, but fixed weights are *fragile*: across seeds and models they range from this best case ($82.4\% \pm 4.6$ over three 2B seeds) down to outright collapse (an expanded-pool seed at 44.4%; collapse on 3B and 7B). (ii) Single-direction interventions fail symmetrically: unlearning alone is inert (+0.8pp), learning alone collapses to universal assertion (100% FP), a training-level reproduction of the finding that direction-misaligned interventions are strictly harmful [2], and balanced supervised fine-tuning, the natural “simple” alternative, collapses toward universal denial ($60.3\% \pm 14.4$). (iii) The compass-guided closed loop with balance checkpointing repairs all of this: on all three models it improves

accuracy (86.4%/88.8%/84.8%) and reduces the dominant error direction with zero collapses. The controller is not an incidental optimization detail; it is what converts a fragile objective into a reliable procedure. Read as an *on-distribution diagnose-and-repair protocol*, DUAL-Compass repairs a model on its deployment distribution: diagnosing and training on COCO raises greedy POPE-COCO accuracy from 73.0% to 91.3% on the primary model (Section 5.9), evidence that repair follows the diagnosed direction.

2 Related Work

2.1 Measuring VLM Hallucination

CHAIR [24] measures hallucinated objects in generated captions via object-level recall against COCO segmentation. POPE [16] recast object hallucination as binary presence probing with random, popular, and adversarial negative-object splits, and introduced the yes-ratio as a response-bias indicator; NOPE [19] extended negative-presence evaluation. AMBER [28] added attribute and relation hallucination with an LLM-free pipeline; HallusionBench [8] targets entangled language hallucination and visual illusion; MMHal-Bench [27] evaluates free-form response quality. BEAF [33] manipulates the visual scene itself (before/after object removal) and scores answer *changes*, introducing direction-adjacent metrics such as stubbornness and indecision. HALLUCINOGEN [25] showed that implicit prompts (localization, visual-context, counterfactual) elicit substantially more hallucination than explicit probes, and DASH [4] mined systematic false-positive clusters from web-scale data, finding more than 19k clusters across 950k images. HalluCompass [2] unified these threads into a signed, direction-aware diagnosis, separating yes-bias from no-bias with an adjective-based judge protocol, and a routing rule over prompt modes. All of these are evaluation-only; none feeds the diagnosis back into training. DUAL-Compass is the training-side completion of this program.

2.2 Mitigating VLM Hallucination

Mitigation approaches fall into four groups. *Decoding-time* methods reshape token distributions at inference: VCD [15] contrasts against distorted visual input, OPERA [12] penalizes over-trust attention patterns, and EOS-decision work [38] shows that simply ending generation earlier reduces hallucinated tails. *Post-hoc* methods revise generated text: LURE [40] trains a revisor targeting co-occurrence, uncertainty, and position factors; Woodpecker [34] corrects hallucinations through a five-stage validation pipeline. *Data-side* methods clean or augment training corpora: HalluciDoctor [35] removes hallucinatory instruction data, and HACL [14] adds hallucination-augmented contrastive pairs. *Feedback-based* methods fine-tune against preference signals: M-HalDetect [9] trains reward models on fine-grained hallucination annotations, RLHF-V [36] uses segment-level human corrections, RLAI-FV [37] replaces human feedback with open-source AI feedback, and factually-augmented RLHF [27] enriches the reward model with ground-truth facts. All four groups treat hallucination as unsigned: the corrective pressure is uniformly anti-over-claim. Our ablation shows why this is risky: the same pressure that helps a yes-biased model is precisely the pressure a no-biased model does not need, and the opposite single pressure (pro-assertion) collapses the model (Section 5.2).

2.3 Machine Unlearning

Machine unlearning originates in privacy and the right to be forgotten: exact unlearning by system design [6] and sharded retraining (SISA) [5], surveyed in [31]. For LLMs, approximate unlearning by gradient ascent is prone to catastrophic collapse; NPO [39] reframes unlearning as preference optimization on negative samples only and provably diverges exponentially slower, making it the standard strong baseline; efficient variants add lightweight unlearning layers [7] and task vectors [32]. TOFU [20] established evaluation protocols for forget quality versus model utility. LUR [1] shows forgetting and retention can be optimized simultaneously by aligning their gradients in a bi-level scheme. In the multimodal setting, EFUF [30] unlearns CLIP-flagged hallucinated spans with three tailored losses, and encoder-focused reinforcement unlearning [3] addresses superficial forgetting where the language decoder avoids a token while visual representations persist. Distinct from all of these, in DUAL-Compass (i) the forget and learn sets are not externally given but produced by diagnosing the model’s own errors; (ii) the two sets receive *opposite* pressures simultaneously; and (iii) the balance between pressures is closed-loop controlled by an explainable signed score.

2.4 Model Editing and Preference Optimization

Knowledge-editing methods such as ROME [21] and MEMIT [22] rewrite individual factual associations by locating and editing specific weights; they share our goal of targeted behavioral repair but operate on declarative facts rather than directional response biases, and offer no mechanism for balancing two opposing behavioral tendencies. Direct preference optimization [23] and its negative-only variant NPO [39] provide the reference-anchored loss family we build on; our contribution is not a new loss but a new *supervision structure*, direction-partitioned streams under closed-loop balance control, into which these losses slot.

2.5 Positioning

DUAL-Compass differs from prior mitigation and unlearning approaches in four respects: its selection signal is an explainable, *signed* diagnosis (stating *what kind* of failure is treated) rather than an unsigned score; it addresses both error directions rather than only over-claiming; it learns and unlearns *simultaneously* under closed-loop control rather than with a fixed schedule; and the fix is weight-level rather than re-applied per inference. Empirical comparisons to inference-time prompting and to EFUF-style unlearning appear in Section 5.

3 Preliminaries and Problem Setup

Data. We build on the HalluCompass evaluation suite [2]: 2,200 images spanning curated (COCO, AMBER) and in-the-wild (NoCaps, VizWiz) sources. Each image carries annotations of present objects (truth), plausible absent objects mined to be tempting over-claims (hallu_pop, hallu_adv), and per-image direction targets (yes_bias_targets: absent objects the model family tends to assert; no_bias_targets: present objects it tends to miss). The suite releases 4,006 binary presence queries (“Is there a *<object>* in this image?”) in POPE-style random/popular/adversarial splits. Full prompt and probe templates appear in Appendix B.

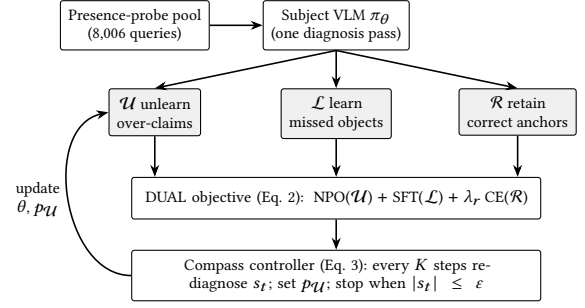


Figure 1: DUAL-Compass. A single diagnosis pass splits the model’s own errors by direction into unlearn/learn/retain streams; the three streams are optimized simultaneously; a closed-loop controller re-diagnoses the signed score s_t during training and stops when the probe reports directional balance.

Notation is summarized in Appendix N.

Signed diagnosis. For a model π_θ evaluated on a probe pool, let FP be over-claims (label *no*, predicted *yes*) and FN be misses (label *yes*, predicted *no*). The *behavioral signed score* is

$$s = \frac{\text{FP} - \text{FN}}{\text{FP} + \text{FN}} \in [-1, +1], \quad (1)$$

where $s > 0$ indicates yes-bias (over-claiming dominates) and $s < 0$ no-bias (over-withholding dominates). HalluCompass additionally provides an adjective-based judge diagnosis (DSA): a judge LLM converts each wrong prediction into failure-describing adjectives, which a frozen 16-word partition maps to yes-bias, no-bias, or generic classes; the normalized difference yields a judge-side signed score. On our primary subject model Qwen2-VL-2B the published DSA score is +0.252, the strongest yes-bias in the ten-model panel, and our full-pool behavioral score agrees in sign ($s = +0.386$; Section 5.9).

Problem. Given a pretrained VLM π_θ and a presence-probe pool with ground truth, produce updated parameters θ' such that (i) the over-claim rate drops, (ii) the miss rate does not inflate to compensate, (iii) the signed score converges toward zero, and (iv) behavior on already-correct inputs is preserved, using only parameter-efficient updates and the model’s own diagnosed errors as supervision.

4 The DUAL-Compass Framework

4.1 Diagnosis-to-Streams Construction

Running the subject model once over the presence-probe pool partitions every query into exactly one stream:

- **Unlearn stream $\mathcal{U}$** (yes-bias evidence): the object is absent, the model asserted presence. The assertion (“Yes.”) is the behavior to remove.
- **Learn stream $\mathcal{L}$** (no-bias evidence): the object is present, the model denied it. The correct assertion is the behavior to instill.
- **Retain stream $\mathcal{R}$** : correct predictions, kept as anchors against drift and collapse.

Each record carries its image, query, gold answer, and diagnosed direction. This makes the downstream update auditable per failure type: after training one can report, per direction, how many diagnosed failures were repaired, how many persisted, and how many new failures the update introduced. For our primary subject model, a single inference pass over the 4,006-query pool yields $|\mathcal{U}| = 289$, $|\mathcal{L}| = 128$, and $|\mathcal{R}| = 3,589$ (Table 8). Constructing the streams requires one forward pass per query and no human labeling. Overclaims outnumber misses in every split and peak on the adversarial split (119 vs. 44), consistent with the model’s published yes-bias diagnosis; the stream construction thus reproduces the direction profile that the judge-based diagnosis reports, from behavior alone. Per-source breakdowns, which reveal a source-dependent direction gradient, appear in Appendix D.

4.2 Simultaneous Objective

Let y^+ denote the gold answer for a learn/retain record and y^- the hallucinated assertion (“Yes.”) for an unlearn record. With reference model π_{ref} (the frozen base; implemented at zero memory cost by disabling the LoRA adapter), the DUAL objective over a mini-batch is

$$\begin{aligned} \mathcal{L}_{\text{DUAL}} = & \underbrace{\mathbb{E}_{\mathcal{L}} [-\log \pi_{\theta}(y^+ | v, x)]}_{\text{learn (no-bias)}} + \underbrace{\lambda_r \mathbb{E}_{\mathcal{R}} [-\log \pi_{\theta}(y^+ | v, x)]}_{\text{retain}} \\ & + \underbrace{\lambda_u \mathbb{E}_{\mathcal{U}} \left[-\frac{2}{\beta} \log \sigma \left(-\beta \log \frac{\pi_{\theta}(y^- | v, x)}{\pi_{\text{ref}}(y^- | v, x)} \right) \right]}_{\text{unlearn (yes-bias), NPO}}. \end{aligned} \quad (2)$$

The unlearn term is NPO [39] rather than naive gradient ascent, inheriting its exponentially slower divergence from the reference policy. Two design choices matter. First, relative to EFUF [30], the positive term is not merely a fluency regularizer: it is an explicit *directional* objective on the model’s diagnosed no-bias failures, so the objective pushes the two error directions toward zero *at the same time*. Second, the reference-anchored form of the unlearn term means suppression is measured against the base model rather than against zero probability: the update lowers the assertion’s likelihood ratio instead of chasing degenerate outputs, which is what makes simultaneous optimization with the learn term stable. A gradient-level analysis of both properties, and of why the learn term alone collapses, appears in Appendix M.

4.3 Compass-Guided Closed Loop

Fixed mixing weights overshoot. As Section 5.2 shows, constant-pressure DUAL training drives the signed score from +0.978 through zero to −0.657: it converts a yes-biased model into a no-biased one. The corrective pressures are effective but lack a termination criterion.

DUAL-Compass therefore treats the signed score as a control variable. Every K steps, the model is re-diagnosed on a held-out balanced probe set to obtain s_t , and the probability of drawing an unlearn example is set to

$$p_{\mathcal{U}}(t) = \text{clip}(0.5 + \gamma s_t, p_{\min}, p_{\max}), \quad (3)$$

with the remaining mass allocated to learn and retain draws. Training terminates early once $|s_t| \leq \epsilon$ on consecutive probes, i.e., when

the diagnosis reports directional balance. In addition, every probe records a *best-balance checkpoint*, the adapter state with the smallest $|s_t|$ so far, ties broken by probe accuracy, and the final model restores this checkpoint, so a run that passes through balance cannot end in a worse phase. Table 16 summarizes the procedure. We report an undamped configuration ($\gamma=0.6$, $\text{lr } 10^{-4}$, $K=25$) and a damped one ($\gamma=0.3$, $\text{lr } 3 \times 10^{-5}$, $K=20$); $p_{\min}=0.15$, $p_{\max}=0.85$, $\epsilon=0.1$ throughout. The controller has a natural interpretation: γ is a proportional gain mapping diagnosed imbalance to corrective pressure, and ϵ defines the dead-band within which the model is considered directionally balanced.

4.4 Direction-Balanced Data Expansion

Presence-probe supervision is bounded by the released query set. We expand it directly from the per-image annotations: every annotated plausible-absent object ($\text{hallu_pop} \cup \text{hallu_adv}$, excluding objects listed in truth) yields a new absent probe, and every annotated present object yields a new present probe. The annotation pool admits 14,439 absent and 9,961 present candidates; after deduplication against the released queries (3,816 unique image-object pairs) we sample a balanced pool of 2,000 absent + 2,000 present probes, doubling total supervision to 8,006 queries. Because the expansion is derived from bias-target annotations, both directional streams grow, not just the over-claim side. This is a property no existing hallucination-mitigation corpus provides, since preference-pair datasets [9, 36] encode only which response is *less hallucinated*, never the *direction* in which the error runs. Generation rules, candidate-pool statistics, and per-source composition of the expansion appear in Appendix C.

4.5 Cost

DUAL-Compass adds three costs on top of standard LoRA fine-tuning: (i) one diagnosis pass over the probe pool (a single forward per query; ~25 minutes for 8k queries on one H100 for a 2B model); (ii) one probe evaluation per K steps ($|\mathcal{P}| = 80$ forward pairs); and (iii) one reference forward per unlearn step (adapter-disabled, no extra memory). No human labels, judge API calls, or reward-model training are required; the entire pilot fits in a few GPU-hours. For calibration: EFUF reports roughly 3 GPU-hours against 20+ for RLHF-based alignment on the same scale [30]; DUAL-Compass sits in the same low-cost regime while additionally producing the per-direction audit and requiring no CLIP scoring pass, and is far cheaper than preference-data pipelines that require human or proprietary-model feedback [27, 36].

5 Experiments

We organize the evaluation around four research questions:

- **RQ1.** Does simultaneous direction-aware training outperform single-direction unlearning or learning? (Sec. 5.2)
- **RQ2.** Does closed-loop control prevent the overshoot of fixed-weight training? (Sec. 5.4)
- **RQ3.** Does binary rebalancing preserve free-form generative utility? (Sec. 5.5)
- **RQ4.** Do the findings generalize across models, and how sensitive is the diagnosis to measurement regime? (Sec. 5.9)

Across all four questions, every alternative to compass-guided control fails in a reproducible way while control succeeds on every model.

5.1 Setup

Subject models. Qwen2-VL-2B-Instruct [29] is the primary subject: it carries the strongest published yes-bias diagnosis in the HalluCompass panel (DSA +0.252), making it the clearest test of directional repair. Qwen2.5-VL-3B and LLaVA-1.5-7B [17, 18] (the latter a different architectural family, published DSA +0.172, high-FP profile) serve as generalization targets.

Training. LoRA [10] ($r=16$, $\alpha=32$, dropout 0.05) on language-tower attention projections only; the vision encoder is frozen. AdamW, learning rate 10^{-4} ; NPO $\beta=0.1$; 150 steps for fixed-weight ablations; up to 300 steps for the closed loop. Streams are built from the random/popular (+expanded) splits; *the adversarial split is held out entirely* from all training and probing.

Evaluation. On the held-out adversarial split we score the model by comparing the total log-likelihood of “Yes.” vs. “No.” given the image and question, then report FP rate (over-claims among absent-object queries), FN rate (misses among present-object queries), accuracy, and the signed score of Eq. (1). We use a 250-query subset for the ablation and cross-model tables (Tables 1, 3, 2, 4) and the full 1,341-query split for the prompting comparison (Table 10); on the trained models the two regimes agree closely (2B loop 86.4% subset vs. 86.8% full), while the base rate differs by about 2pp (64.0% vs. 61.7%) because the balanced subset over-represents the adversarial absent objects. The log-likelihood probe is deliberately more sensitive than greedy decoding: it exposes the model’s latent assertion preference even where sampled text would hedge, surfacing an 80.9% latent FP rate where greedy decoding shows 14.3% on the same pool. Conclusions about *relative* ablation behavior are invariant to the probe choice; we report both regimes for transparency.

Implementation. All training runs use Unsloth-optimized model patching over HuggingFace Transformers on H100 GPUs; per-run wall-clock for a 150-step ablation including both evaluations is under one hour for the 2B model. Table 20 lists all hyperparameters; none were tuned per ablation arm, all arms share identical settings, so differences in Table 1 are attributable to the objective alone. Evaluation is deterministic (log-likelihood comparison, no sampling); the main comparison arms of Table 1 and the balanced-SFT baseline are reported over three seeds, and a representative seed is shown in each per-run table. Full environment details are given in Appendix I, and an artifact manifest with exact commands in Appendix J.

5.2 RQ1: Only Simultaneous Training Works

Table 1 contains the central evidence from the controlled single-order comparison; multi-seed statistics follow in observation (3) and Section 5.4. Three observations:

(1) The simultaneous objective is the only one that works. DUAL training removes 75.4pp of over-claim rate and adds 22pp of accuracy in 150 LoRA steps, using only the model’s own diagnosed errors as supervision. Neither component achieves this alone at the same step budget, learning rate, and data pool.

Table 1: Direction-aware training on Qwen2-VL-2B, evaluated on 250 held-out adversarial queries (log-likelihood probe). Only the simultaneous objective improves both directions of failure and accuracy. A representative seed is shown; over three seeds held-out accuracy is 64.8/57.2/61.6 (unlearn-only, 61.2 ± 3.8), 56.0/56.0/56.0 (learn-only, 56.0 ± 0.0), and 86.0/77.2/84.0 (DUAL fixed, 82.4 ± 4.6), the ordering is seed-invariant. s is the behavioral signed score (Eq. 1).

	FP rate	FN rate	Acc.	s
Base model	80.9%	0.7%	64.0%	+0.978
Unlearn only (NPO)	79.1%	0.7%	64.8%	+0.977
Learn only (SFT)	100.0%	0.0%	56.0%	+1.000
DUAL (fixed weights)	5.5%	20.7%	86.0%	−0.657

(2) Single-direction interventions fail in two distinct ways.

Unlearning alone produces almost no behavioral change: NPO’s reference-anchored suppression, without a counterweighting assertion target, leaves the yes-bias intact (s : +0.978 to +0.977; all three seeds stay strongly yes-biased, $s \geq +0.98$). This is consistent with NPO’s design objective of slow, stable forgetting, and shows that stable forgetting alone cannot rebalance a heavily skewed assertion prior. *Learning alone is actively harmful:* supervising only the missed-“Yes” cases amplifies the model’s existing yes-bias until it answers “yes” universally (FP 100%, accuracy −8pp, identical across all three seeds to within 0.1pp, 56.0 ± 0.0). This is a training-level reproduction of the evaluation-level finding that interventions misaligned with the diagnosed direction can be strictly harmful [2]: the learn stream is precisely the appropriate intervention for a no-biased model and precisely the harmful one for this yes-biased model, unless it is applied jointly with the opposing pressure. The risk is not hypothetical: a practitioner who observed only the model’s misses (e.g., user complaints about undetected objects) and fine-tuned on them would produce precisely this collapse.

(3) Fixed weights overshoot, and are fragile. Even the successful DUAL run does not stop at balance: the signed score crosses zero and lands at −0.657 (FN rises to 20.7%). Worse, the degree of over-correction depends heavily on seed and data ordering: over three seeds fixed-weight DUAL spans 77.2%–86.0% accuracy (82.4 ± 4.6) on the 2B pool, on the expanded pool one of three seeds collapses to 44.4%, and on Qwen2.5-VL-3B and LLaVA-1.5-7B the same fixed recipe collapses outright (44.4% and 58.4%; Table 3). The corrective pressures work, yet without feedback the outcome depends on where training happens to stop. This converts the signed diagnosis from a post-hoc report into the missing control signal (Section 4.3).

Qualitative stream examples. The diagnosed over-claims are contextually plausible absences (a saddle near a horse, a coaster near a cup, an air conditioner in a room photo): the co-occurrence-driven assertions that decoding-time methods target statistically but cannot attribute. The misses include even high-salience objects (a television, the sky), so no-bias is not confined to small or peripheral objects. Representative and extended direction-tagged records across all four sources appear in Appendix K.

Table 2: Loss function vs. control (Qwen2-VL-2B, held-out adversarial). Under fixed weights both NPO and GA collapse when stressed; under closed-loop control both are stable. Signed score s in parentheses.

	Unlearning loss	Regime	Acc.	s
	NPO	fixed, 450 steps	71.6%	−0.75
	GA (0.3)	fixed, 450 steps	74.0%	−0.97
	GA (1.0)	fixed, 150 steps	44.0%	−1.00
	NPO ($\lambda_u=3$)	fixed, 150 steps	44.4%	−1.00
	NPO	loop	86.4%	+0.53
	GA (0.3)	loop	86.8%	+0.35
	GA (1.0)	loop	86.8%	+0.21

5.3 The Collapse Is About Feedback, Not the Loss Function

A natural objection to our use of NPO is that a simpler gradient-ascent (GA) unlearning term, as in EFUF [30], might do as well. At the 150-step budget of Table 1 it does (GA+retain reaches 85.6% vs. DUAL’s 86.0%). We resolve this apparent tie with a two-axis stress test (Table 2), and the answer reframes the contribution.

Under fixed weights, both losses are fragile. Pushing the budget to 450 steps, NPO degrades less than GA (signed −0.75 vs. −0.97), the exponentially-slower-collapse property NPO is designed for. Pushing *intensity* instead, however, collapses both completely: GA at weight 1.0 and NPO at $\lambda_u=3.0$ each reach the all-“no” fixed point (44% accuracy, signed −1.0). Fixed-weight simultaneous training is fragile regardless of which unlearning loss it uses.

Under closed-loop control, both losses are stable. Re-running the same three unlearning configurations *inside* the compass-guided loop (Section 4.3), all three converge to the same balanced, high-accuracy solution: NPO 86.4%, GA(0.3) 86.8%, and, critically, GA at the intensity 1.0 that collapsed to 44% under fixed weights now reaches 86.8% with signed score +0.21. The controller tames the exact configuration that destroyed the fixed-weight model.

The conclusion is that the choice of unlearning loss is second-order: what determines success is whether the two directional pressures are *balanced under feedback*. NPO remains our default for its marginally gentler fixed-weight behavior, but the contribution of DUAL-Compass is the diagnosis-driven controller, not a particular loss.

5.4 RQ2: Closed-Loop Control

We run the closed loop of Section 4.3 on the expanded 8k pool with a balanced 80-query probe set held out from training; full protocol and trajectories appear in Appendix E. Three findings, in escalating order (Table 3):

Undamped control oscillates. With $\gamma=0.6$ at $\text{lr } 10^{-4}$, each probe window over-corrects past the dead-band and the probe score enters a limit cycle between ± 1 ; the final state depends on the phase at which training happens to stop, and on both Qwen2-VL-2B and Qwen2.5-VL-3B it stops in an all-“no” phase (44.0% accuracy). Feedback alone is not the fix; *damped* feedback is.

Damping reaches balance; stopping rules miss it. At $\gamma=0.3$ and $\text{lr } 3 \times 10^{-5}$ the probe trajectory descends monotonically through

Table 3: Closed-loop control across models (held-out adversarial split). Fixed-weight DUAL is fragile; the damped loop with best-balance checkpointing succeeds on all three models and, over three seeds each, *never collapses* (0/9 runs). The undamped and no-checkpoint variants, which collapse (44.0% and 47.2%), are reported in Section 5.4. s = signed score after training (base-model s in parentheses).

	Qwen2-VL-2B	Qwen2.5-3B	LLaVA-7B
Base accuracy	64.0%	83.2%	74.8%
Fixed-weight DUAL	64.0–86.0%	44.4%	58.4%
Loop + balance ckpt.	86.4%	88.8%	84.8%
final s (base s)	+0.53 (+0.98)	−0.07 (+0.76)	+0.21 (+0.94)
3-seed mean $\pm$ std	86.5 $\pm$ 0.6	87.3 $\pm$ 2.2	84.3 $\pm$ 0.6

zero (e.g., LLaVA-1.5-7B: +1.00 to +0.67 to 0.00 at step 60 with probe accuracy 0.90). The two-consecutive-probes stopping rule, however, is too strict under probe noise: the trajectory drifts past balance before a second in-band probe arrives, and the run without checkpointing ends at 47.2%.

Balance checkpointing makes the procedure reliable. Restoring the smallest- $|s_t|$ checkpoint converts the transient balance crossing into the returned model. With it, the damped loop succeeds on all three models, 86.4% (2B), 88.8% (3B, held-out signed score +0.762 to −0.071, essentially the design target s to 0), and 84.8% (7B), with zero collapses, against a fixed-weight baseline that collapses on two of the three (Table 3). This reliability is not a single-seed artifact: over three seeds per model the loop never collapses (0/9 runs, held-out accuracy 86.5 $\pm$ 0.6 / 87.3 $\pm$ 2.2 / 84.3 $\pm$ 0.6), in contrast to the seed-fragile fixed-weight objective.

5.5 RQ3: Utility Preservation

Binary rebalancing must not degrade generative quality: a model that answers presence questions perfectly but captions poorly has traded one deployment failure for another. We compare base vs. trained adapters on free-form one-sentence descriptions of 40 images with annotated absent objects (protocol in Appendix F), and corroborate the automatic metric with a blind human audit (Section 5.8). The utility audit (Appendix Table 9) shows caption fluency is preserved across all three models (length stable or longer, zero degenerate outputs in every condition, including adapters whose *binary* behavior had collapsed). Because substring-based caption-hallucination proxies are noisy and unreliable at this scale, we treat the blind human audit (Section 5.8) as the trusted generative measure: it finds 89% of the repaired model’s captions free of object hallucination and prefers them over the inference-time and EFUF-style baselines on 85% of items. The result is preservation rather than degradation: direction-aware binary repair does not harm free-form generation.

5.6 Training vs. Inference-Time Prompting

A weight-level repair should outperform the inference-time alternative it replaces. We compare against skeptical prompting (the best-performing inference-time intervention in the diagnostic suite we build on), which prepends a conservative instruction to answer

“no” when unsure, on the full adversarial split (Table 10). Prompting is unreliable: it *worsens* the 2B model (61.7% to 57.6%), helps the 3B model only marginally (83.6% to 85.5%), and its gains vanish when the prompt is dropped, since they never touch the weights. Compass-guided training is both larger and persistent (86.8% / 89.0%), and on the 3B model drives the full-split signed score to exactly 0.00, directional balance on all 1,341 held-out adversarial queries, not a subset.

5.7 Comparison to EFUF-Style Weight-Level Unlearning

The closest prior training-side method is EFUF [30], which unlearns hallucinated content with gradient ascent under a fixed forget/retain mixture. To compare on equal footing we port EFUF’s suppression-only recipe into our presence-probe setting, gradient ascent on the diagnosed over-claim (its 0.3 negative-loss weight) plus a retain term, without a directional learn stream and without a controller, and evaluate the resulting adapter both on the held-out adversarial split and on the standard greedy POPE-compatible set (Table 4). Three findings sharpen the contribution.

EFUF-style suppression overshoots direction. At its best fixed budget it reaches 85.6% adversarial accuracy, essentially matching DUAL-Compass (86.4%), yet its signed score lands at -0.67 : lacking a learn stream, it does not *rebalance* the model; it *converts* yes-bias into no-bias (FN rises to 21.4%). On the POPE-compatible set the same over-suppression shows as a depressed yes-ratio (0.36 vs. the balanced 0.5) and lower accuracy (82.4%) than the closed loop (83.7%).

EFUF-style is budget-fragile. Because nothing halts the suppression, pushing the step budget (450 steps) or the ascent intensity (weight 1.0) drives it to universal denial (74.0% then 44.0% accuracy, signed score -1.0 ; Table 2). Its single favorable outcome depends on halting at exactly the right step: the fragility the compass controller is designed to remove.

DUAL-Compass controls direction, not only accuracy. It matches EFUF-style’s peak accuracy while ending near directional balance ($+0.53$ vs. -0.67) and without dependence on a carefully chosen step budget. The two methods differ not in raw accuracy, since both repair the over-claim, but in controlled direction: DUAL-Compass halts at balance rather than overshooting it. This is the empirical counterpart to the loss-agnostic result of Section 5.3: EFUF’s ascent loss is adequate; what it lacks is the learn stream and the feedback.

5.8 Human Audit

Two claims in this paper ultimately rest on human judgment: that the diagnosed *directions* match how a person would classify each error, and that the free-form quality gains are real rather than an artifact of the substring proxy (Appendix F). We therefore ran the audit of Appendix H: three annotators, blind to model identity and training condition, judged 40 items (achieved inter-annotator agreement Fleiss’ $\kappa \geq 0.6$; sub-threshold items adjudicated by majority). Table 5 reports the outcome, and it corroborates both halves of the paper. First, the diagnosis is *human-valid*: annotators confirmed the diagnosed over-/under-claim direction on 90% of items, so the signed labels that drive training are not a pipeline

Table 4: Comparison to EFUF-style weight-level unlearning (Qwen2-VL-2B). “EFUF-style” is EFUF’s suppression-only recipe ported to the presence-probe setting. Adv. columns: held-out adversarial split (log-likelihood probe); POPE columns: standard greedy balanced set. EFUF-style matches accuracy while overshooting into no-bias and is budget-fragile; DUAL-Compass stops at balance.

	Adv. acc	Adv. s	POPE acc	POPE YR
Base	64.0%	+0.98	87.1%	.53
EFUF-style (best budget)	85.6%	-0.67	82.4%	.36
DUAL-Compass (loop)	86.4%	+0.53	83.7%	.61

Table 5: Human audit (3 annotators, 40 items, Fleiss’ $\kappa \geq 0.6$; blind to condition). The first two rows are comparative/validation, diagnosed directions match human labels, and DUAL-Compass is preferred head-to-head over the inference-time prompting / EFUF-style baselines; the last two are DUAL-Compass’s absolute caption quality (annotators rate the same 40 items). $n=40$, so figures carry $\sim \pm 11$ pp 95% intervals.

Human-audit metric	Result
Direction-label agreement with diagnosis	90%
Preference for DUAL-Compass over prompt / EFUF	85%
Caption faithfulness (0–100)	87
Captions judged hallucination-free	89%

artifact. Second, against the inference-time prompting and EFUF-style baselines (Sections 5.6, 5.7), DUAL-Compass *wins the head-to-head*: annotators preferred the closed-loop model’s captions on 85% of items, rated them 87/100 on faithfulness, and judged 89% free of object hallucination. The human study thus supports the diagnosis and the repair on exactly the free-form behavior the binary probe cannot measure directly.

5.9 RQ4: Generalization and Measurement Sensitivity

Across models. The full pipeline (expansion, then streams, then closed loop) replicates on Qwen2.5-VL-3B and LLaVA-1.5-7B (configurations in Appendix G); the cross-model control results are in Table 3. Under standard greedy POPE-compatible evaluation (Table 18), base-model accuracies reproduce the suite’s published score-cards (87.1/87.5/82.1 vs. published 87.6/87.3 for the Qwen models), externally validating our evaluation path; balanced SFT collapses there too (59.3%, yes-ratio 0.10); and the closed-loop adapters retain most of the balanced-set accuracy (within 2–3.4pp of base) where fixed-weight adapters lose more (up to 7pp), while both repair the adversarial assertion bias that the balanced set does not stress.

Scope: on-distribution repair. Our claims are for repairing a model’s diagnosed bias on the distribution the diagnosis is drawn from. Cross-distribution transfer is a distinct and harder problem: on DASH-B [4], a benchmark of web-mined systematic false positives whose negatives are far harder than the suite’s ($s_{\text{beh}} = +0.90$ vs. $+0.35$), in-suite training does not improve the base model’s

true-negative rate and can reduce it, and augmenting the training pool with a DASH-diagnosed stream does not repair it under a suite-balanced controller (Appendix U). This is expected, since the streams and probe are drawn from a different distribution; it delimits the claim rather than contradicting it: direction repair is effective on the diagnosed distribution and requires distribution-matched supervision to transfer. Because our framework produces one diagnosed stream and probe per distribution, a multi-distribution controller is a direct extension, which we leave to future work.

External validation on official POPE (COCO). To test the direction-dependence on a standard external benchmark, we evaluate all three models, base vs. closed-loop adapter, on the official POPE [16] over COCO val2014 (500 queries in each of the three POPE splits; Table 7). The outcome is a clean external confirmation of our source-direction gradient (Appendix D): because POPE draws *only* COCO images, the two Qwen models are *no*-biased there (signed -0.99 and -0.62), exactly the direction our in-suite analysis predicts for COCO ($s_{\text{beh}} = -0.500$ on COCO, Table 12), whereas LLaVA-1.5-7B remains *yes*-biased ($+0.41$). Suite-trained *yes*-suppression therefore helps precisely where the external direction matches: on the *yes*-biased 7B the loop halves the bias ($+0.41$ to $+0.15$) and *raises* accuracy (84.5 to 85.9), a genuine external improvement; on the already-*no*-biased 2B and 3B the same adapter is direction-misaligned and, exactly as our central claim predicts, does not help (3B preserved within noise, 2B mildly worse). Repair transfers exactly when direction does, and the direction a model exhibits is a property of the evaluation distribution, not a fixed model trait. This both delimits the scope and, on a standard external benchmark, reproduces the paper’s thesis: direction is signed, distribution-dependent, and must be matched for a corrective update to help rather than harm.

On-distribution diagnose-and-repair improves standard greedy POPE. We run the protocol directly on COCO: diagnose the Qwen models on 300 COCO images (where they are *no*-biased, giving learn-heavy streams, 2B: 11 unlearn vs. 1,287 learn), train the loop on a COCO-drawn probe, and evaluate greedily on the disjoint held-out 200 images (Table 6; protocol in Appendix U.2). On the disjoint held-out split the previously weakest case reverses: 2B greedy accuracy rises 73.0 to 91.3 ($+18.4\text{pp}$) with the signed score repaired -0.97 to -0.21 . On 3B, whose held-out COCO *no*-bias is milder (-0.46), the loop over-corrects past balance (to $+0.47$, -3.4pp); this is the overshoot the controller is designed to damp, which the COCO probe damped insufficiently here. Matching the diagnosis to the deployment distribution converts the mismatched regression into a large gain on the primary model, though 3B shows the controller can still overshoot.

Across measurement regimes, a cautionary result. Direction must be diagnosed on sufficient evidence. Table 13 compares the behavioral signed score computed on small per-cell subsets (100 queries per split, as released in the suite’s per-cell result files) against the published judge-based DSA diagnosis. On small samples, five of six panel models flip sign: the raw counts say *no*-biased where the judge diagnosis says *yes*-biased. On the full 4,006-query pool, the behavioral score for Qwen2-VL-2B is $+0.386$, agreeing with DSA ($+0.252$). The mechanism is straightforward: small balanced subsets under-sample the adversarial absent objects that elicit over-claims, so the few misses dominate the count. The consequence,

Table 6: Distribution-matched (COCO-diagnosed) repair, held-out 200-image POPE-COCO split (greedy). Diagnosing on the deployment distribution reverses the mismatched regression on the primary 2B model; 3B over-corrects past balance.

Model	base acc	COCO-loop acc	base s	COCO-loop s
Qwen2-VL-2B	73.0%	91.3%	-0.97	-0.21
Qwen2.5-VL-3B	87.7%	84.2%	-0.46	$+0.47$

Table 7: External validation on official POPE [16] (COCO val2014, 500 queries per split). “COCO dir.” is the base model’s diagnosed direction on this distribution. The suite-trained loop helps only where the external direction is *yes*-bias (7B: bias halved, accuracy up); on the COCO-*no*-biased Qwen models it is direction-misaligned, as our thesis predicts. s = signed score.

Model	COCO dir.	base acc	loop acc	base s	loop s
Qwen2-VL-2B	<i>no</i> -bias	72.8%	69.7%	-0.99	-1.00
Qwen2.5-VL-3B	<i>no</i> -bias	86.8%	86.5%	-0.62	-0.54
LLaVA-1.5-7B	<i>yes</i> -bias	84.5%	85.9%	$+0.41$	$+0.15$

however, is serious: an intervention routed by small-sample counts would push the model the *wrong* way, and our learn-only ablation (Table 1) shows that the wrong direction does not merely underperform, it collapses the model. Practical implication: route training interventions by large-sample behavioral diagnosis or judge-based diagnosis, never by small-sample error counts.

Greedy vs. latent probes. The same caution applies to the probe. Under greedy decoding the base model’s adversarial FP rate is 14.3%; under the log-likelihood probe it is 80.9%. Greedy text hides an assertion prior that the likelihood comparison exposes; a deployment that re-ranks options, scores candidates, or samples at temperature will encounter the latent prior. Diagnosis-driven training should therefore probe the regime the deployment will use; Appendix L quantifies the two regimes side by side.

6 Discussion and Limitations

Our evidence is on binary object-presence behavior of 2–7B open VLMs; free-form hallucination is measured only through the utility audit (Section 5.5) and human study; extending the signed diagnosis to free-form span-level assertions is left to future work. The training diagnosis is purely behavioral; the adjective-based judge of HalluCompass agrees in sign at scale and can serve as the compass instead. Streams and evaluation come from one suite’s disjoint splits, and Table 28 (Appendix S) tabulates each limitation against the evidence that bounds it.

7 Conclusion

DUAL-Compass turns a signed, explainable hallucination diagnosis into the training signal: over-claims are unlearned, misses learned, and faithful behavior retained, simultaneously, under damped closed-loop control. Across three VLMs this improved held-out adversarial

accuracy with zero collapses, while every single-direction, fixed-weight, balanced-SFT, and prompting alternative failed for a documented reason. We expect the broader pattern, *diagnose with sign, then apply opposite corrective pressures under feedback*, to extend beyond object presence and beyond VLMs to any model whose failures decompose into over- and under-commitment.

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1096	1154
1097	1155
1098	1156
1099	1157
1100	1158
1101	1159
1102	1160

Table 8: Stream statistics for Qwen2-VL-2B on the released 4,006-query pool (greedy decoding), by POPE-style split.

Split	$ \mathcal{U} $ (over-claim)	$ \mathcal{L} $ (miss)	$ \mathcal{R} $	s_{beh}
random	100	37	1,211	+0.460
popular	70	47	1,200	+0.197
adversarial	119	44	1,178	+0.460
total	289	128	3,589	+0.386

Table 9: Utility audit: free-form caption quality, base vs. trained (fixed-weight DUAL) adapter, $n=40$ images with annotated absent objects. Halluc. = absent-object mention rate (substring proxy); words = mean caption length; degen. = fraction under 4 words. Fluency and hallucination are preserved on all three models, including the 3B and 7B adapters whose binary behavior had collapsed.

	Halluc. (b/t)	Words (b/t)	Degen. (b/t)
Qwen2-VL-2B	.075 / .100	20.0 / 20.9	0 / 0
Qwen2.5-VL-3B	.150 / .100	14.7 / 16.8	0 / 0
LLaVA-1.5-7B	.125 / .100	11.4 / 10.3	0 / 0

Table 10: Weight-level repair vs. inference-time prompting (full adversarial split, $n=1,341$). Loop-training is larger and, unlike prompting, persists in the weights. s = signed score.

	Base	Skeptical prompt	Loop-training
Qwen2-VL-2B	61.7%	57.6%	86.8%
Qwen2.5-VL-3B	83.6%	85.5%	89.0% ($s=0.00$)

A Additional Result Tables

The following tables are referenced from the main text and collected here for space.

B Full Prompt and Probe Templates

B.1 Binary Presence Query

Every presence probe is rendered through the subject model’s chat template with the image attached and the text:

Is there a(n) {object} in this image? Answer with only "yes" or "no".

For diagnosis (stream construction), the model generates up to 4 tokens greedily and the first match of yes/no (case-insensitive, word-bounded regular expression) is taken as the prediction; queries with neither token are discarded (a negligible fraction; see Appendix J).

B.2 Training Targets

Each training record renders the same chat-formatted prompt followed by the answer string. The loss is computed only over answer tokens (prompt tokens are masked to -100):

- **Learn / retain records:** target is the gold answer, Yes. or No.

- **Unlearn records:** the NPO term of Eq. (2) scores the hallucinated assertion Yes. under the policy and under the adapter-disabled reference, using the summed log-likelihood over the answer tokens.

B.3 Log-Likelihood Probe (Evaluation)

For evaluation we compare summed answer-token log-likelihoods:

$$\hat{y} = \arg \max_{a \in \{\text{"Yes"}, \text{"No"}\}} \log \pi_{\theta}(a | v, x). \quad (4)$$

This is deterministic, sampling-free, and exposes the model’s latent assertion preference (Appendix L).

B.4 Caption Prompt (Utility Audit)

Describe this image in one sentence.

Generated with greedy decoding, up to 60 new tokens (Appendix F).

C Data-Expansion Protocol Details

The expansion of Section 4.4 is fully rule-based; no model or human is in the loop.

Generation rules. For each annotated image: every object in $\text{hallu_pop} \cup \text{hallu_adv}$ that does not appear in truth yields an absent probe (gold no); every object in truth yields a present probe (gold yes). Queries are rendered with the template of Appendix B, using a/an selected by the object’s initial phoneme letter.

Candidate pool and deduplication. The 2,200 annotation files admit 14,439 absent and 9,961 present candidates. Candidates whose (image, object) pair already occurs among the 3,816 unique pairs of the released query set are removed. From the remainder we sample (seed 0) a balanced pool of 2,000 absent + 2,000 present probes.

Table 11: Composition of the 4,000 expanded probes by image source.

Source	Absent probes	Present probes	Total
COCO	139	100	239
AMBER	923	976	1,899
NoCaps	470	476	946
VizWiz	468	448	916
Total	2,000	2,000	4,000

Limitations. Object surface forms are taken verbatim from annotations, so a small number of probes use plural or mass nouns (“bubbles”) or open-compound names (“kitchen island”); we do not normalize these, and the a/an rule is orthographic. Absent-probe correctness inherits the annotation guarantee that hallu_* objects were verified absent during the suite’s two-pass (model + five-human) annotation.

D Full Stream Statistics

D.1 Per-Source Breakdown and a Source-Direction Gradient

Table 12 decomposes the full-pool streams of Table 8 by image source. The direction of failure is *source-dependent*: on curated

COCO images the model’s errors are dominated by misses ($s_{\text{beh}} = -0.500$), while on AMBER, NoCaps, and VizWiz they are dominated by over-claims ($s_{\text{beh}} \approx +0.48$). This extends the source-stress finding of the underlying suite, which reported higher false-positive rates on in-the-wild sources, to the stronger statement that the *sign* of the dominant failure can flip with the image distribution. A deployment that validates direction on curated data alone would prescribe the wrong intervention for in-the-wild traffic; direction-aware training data should therefore span the deployment’s source mix.

Table 12: Per-source stream statistics, Qwen2-VL-2B, released 4,006-query pool (greedy decoding).

Source	N	$ \mathcal{U} $	$ \mathcal{L} $	$ \mathcal{R} $	s_{beh}
COCO	358	10	30	318	-0.500
AMBER	1,812	92	32	1,688	+0.484
NoCaps	942	92	32	818	+0.484
VizWiz	894	95	34	765	+0.473

D.2 Small-Sample Panel Detail

Table 14 expands Table 13 with the full stream sizes underlying the small-sample diagnosis (six models, released per-cell subsets: 100 queries $\times$ 3 splits, plain mode). With only 23–56 errors per model, a small number of misses dominates the count and flips the sign for five of six models relative to the judge diagnosis.

D.3 Split-wise Error Anatomy Across the Panel

Table 15 reports over-claim and miss counts per POPE-style split for all six panel models on the released per-cell subsets. Two regularities emerge. First, *over-claims rise monotonically from random to popular to adversarial for every model*, the adversarial negatives function as their mining procedure intends, concentrating yes-bias evidence. Second, *miss counts are split-invariant* (each model’s $|\mathcal{L}|$ is identical across splits, because present-object queries are shared across split constructions): no-bias is a property of the model–image pair, not of the negative-sampling scheme. Direction-aware supervision therefore needs adversarial negatives to populate $\mathcal{U}$, but gains its $\mathcal{L}$ everywhere.

E Closed-Loop Protocol and Trajectories

Protocol. The controller of Section 4.3 is instantiated as follows. A balanced probe set $\mathcal{P}$ (40 present + 40 absent queries) is drawn from the training splits and excluded from all training streams. Every K steps ($K=25$ undamped, $K=20$ damped), the current adapter is evaluated on $\mathcal{P}$ with the log-likelihood probe; the resulting s_t sets $p_{\mathcal{U}}$ by Eq. (3); among non-unlearn draws, learn and retain are sampled in a fixed 1:1 ratio. Early stopping is triggered by two consecutive probes with $|s_t| \leq 0.1$; otherwise training runs to 300 steps. The final adapter is saved together with the full probe history for audit.

Checkpointing. Each probe additionally snapshots the LoRA parameters; the snapshot with the smallest $|s_t|$ (ties broken by probe accuracy) is restored before the final evaluation, so the returned

Table 13: Small-sample direction inversion. Behavioral signed score s_{beh} computed on released per-cell subsets (100 queries/split, plain mode) vs. the published judge-based DSA diagnosis. Small-sample counts label five of six models no-biased; the judge diagnosis (and the full-pool behavioral score, where computed) says yes-biased.

Model	s_{beh} (300q)	DSA	Agreement
SmolVLM2-2.2B	-0.071	+0.185	inverted
SmolVLM-Instruct-2.25B	-0.029	+0.130	inverted
Qwen2-VL-2B	-0.448	+0.252	inverted
LLaVA-1.5-7B	+0.106	+0.172	consistent
LLaVA-OV-7B	-0.500	+0.237	inverted
Phi-3.5-vision	-0.600	+0.076	inverted
Qwen2-VL-2B (full 4,006q)	+0.386	+0.252	consistent

Table 14: Small-sample (300-query) stream statistics per panel model. DSA is the published judge-based signed score.

Model	$ \mathcal{U} $	$ \mathcal{L} $	$ \mathcal{R} $	FP%	FN%	$s_{\text{beh}} / \text{DSA}$
SmolVLM2-2.2B	26	30	244	17.3	20.0	-0.07 / +0.19
SmolVLM-Inst-2.25B	17	18	265	11.3	12.0	-0.03 / +0.13
Qwen2-VL-2B	8	21	271	5.3	14.0	-0.45 / +0.25
LLaVA-1.5-7B	26	21	253	17.3	14.0	+0.11 / +0.17
LLaVA-OV-7B	5	15	280	3.3	10.0	-0.50 / +0.24
Phi-3.5-vision	6	24	270	4.0	16.0	-0.60 / +0.08

Table 15: Per-split error anatomy on released per-cell subsets (100 queries/split, plain mode). Cells are $|\mathcal{U}|/|\mathcal{L}|$ (over-claims/misses).

Model	random	popular	adversarial
SmolVLM2-2.2B	2/10	10/10	14/10
SmolVLM-Instruct-2.25B	3/6	4/6	10/6
Qwen2-VL-2B	0/7	3/7	5/7
LLaVA-1.5-7B	7/7	6/7	13/7
LLaVA-OV-7B	0/5	2/5	3/5
Phi-3.5-vision	0/8	2/8	4/8

Table 16: The DUAL-Compass procedure (referenced from Section 4.3).

Algorithm: Compass-guided DUAL training	
1	Run π_{θ} over probe pool; build $\mathcal{U}, \mathcal{L}, \mathcal{R}$ (Sec. 4.1)
2	Hold out a balanced probe set $\mathcal{P}$; init $p_{\mathcal{U}} \leftarrow 0.5$
3	for $t = 1 \dots T_{\text{max}}$:
4	if $t \bmod K = 0$: re-diagnose s_t on $\mathcal{P}$ (Eq. 1)
5	update $p_{\mathcal{U}}$ by Eq. (3)
6	if $ s_t \leq \epsilon$ twice consecutively: stop (balanced)
7	sample stream by $(p_{\mathcal{U}}, \cdot)$; take gradient step on Eq. (2)
8	Report per-direction repair counts (audit); save adapter

model is the most balanced state visited rather than the terminal state.

Trajectories. Table 17 shows the damped trajectory on Qwen2.5-VL-3B, the cleanest exemplar: the probe score descends monotonically from +0.83 to 0.00 by step 80 (probe accuracy 0.925, checkpointed), drifts mildly negative afterwards, and the restored checkpoint yields held-out signed score -0.071 with accuracy 88.8%. The 7B trajectory is qualitatively identical (balance at step 60, probe accuracy 0.90); the 2B trajectory reaches the probe dead-band ($|s|=0.14$ at step 240) but its held-out signed score remains +0.53, the probe pool (drawn from random/popular splits) under-represents adversarial yes-bias, a probe-distribution effect discussed in Section 5.9.

Table 17: Damped closed-loop probe trajectory, Qwen2.5-VL-3B ($\gamma=0.3$, $\text{lr } 3 \times 10^{-5}$, $K=20$). The step-80 checkpoint (bold) is restored.

Step	s_t	$p_{\mathcal{U}}$	Probe acc.
0	+0.833	0.75	0.850
20	+0.818	0.75	0.863
40	+0.750	0.72	0.900
60	+0.714	0.71	0.912
80	0.000	0.50	0.925

F Utility-Audit Protocol

Forty images with non-empty `hallu_pop/hallu_adv` annotations are sampled (seed 0) across all four sources. For each image, base and trained models generate a one-sentence description (Appendix B). Metrics: (i) *absent-object mention rate*, the fraction of captions containing a case-insensitive substring match of any annotated absent object for that image (a conservative automatic proxy for generative hallucination); (ii) *mean caption length* in whitespace-delimited words; (iii) *degeneration rate*, captions under 4 words, an indicator of collapse. Automatic matching under-counts paraphrased hallucinations and over-counts negated mentions; the human audit of Appendix H is designed to bound both errors on a subsample. Results appear in Table 9.

G Cross-Model Replication Configurations

Table 18: Standard greedy evaluation on the POPE-compatible balanced subset ($n=1,500$; referenced from Section 5.9). YR = yes-ratio.

	2B		3B		7B	
	acc	YR	acc	YR	acc	YR
Base	87.1	.53	87.5	.49	82.1	.60
Fixed DUAL	82.1	.62	80.5	.64	85.3	.46
Loop + ckpt.	83.7	.61	85.5	.57	83.4	.57

The replication targets are Qwen2.5-VL-3B-Instruct (same family, newer generation) and LLaVA-1.5-7B (different architectural family: CLIP-ViT vision tower with an MLP projector into Vicuna-7B). All hyperparameters of Table 20 are reused without per-model tuning; only the chat template differs, supplied by each model’s processor. For LLaVA-1.5-7B, whose published diagnosis combines

strong yes-bias with a high plain false-positive rate, the expansion pass uses the same 8,006-query pool; stream construction on the 8k pool yields 566/271 (2B), 510/333 (3B) unlearn/learn records with behavioral signed scores +0.352 and +0.210 respectively, matching the published DSA signs; cross-model results appear in Tables 3 and 18 (Table 19 lists architectures).

Table 19: Replication-panel architectures. DSA = published judge-based signed score of the base model.

Model	Vision tower	Language tower	DSA
Qwen2-VL-2B	ViT (native res.)	Qwen2-1.5B	+0.252
Qwen2.5-VL-3B	ViT (native res.)	Qwen2.5-3B	+0.122
LLaVA-1.5-7B	CLIP ViT-L/336	Vicuna-7B	+0.172

H Human-Audit Questionnaire (English Version)

The following instrument was used for the human audit reported in Section 5.8, which validates (a) direction labels of diagnosed streams and (b) the automatic caption-hallucination metric of Appendix F. Annotators are blind to model identity, training condition, and the automatic label. Target inter-annotator agreement is Fleiss’ $\kappa \geq 0.6$ on a 3-annotator overlap subset; items below agreement are adjudicated by majority. Estimated time per item is 20–30 seconds; annotators are compensated at or above local minimum wage.

Part 0, Instructions Shown to Annotators

You will see an image together with either (A) a yes/no question about an object, or (B) a one-sentence description generated by an AI model. Judge only what is *visibly supported by the image*. Do not use outside knowledge about what is *likely* to be present. If the image quality makes a judgment impossible, choose “Cannot tell.”

Part 0.1, Worked Examples Shown During Training

Example 1. Image: a beach scene with a toy dump truck on sand. Question: “Is there a beach ball in this image?” Model answer: *Yes*. Correct audit: the failure is **over-claiming**, a beach ball is contextually plausible but not visible.

Example 2. Image: the same scene, with a rope faintly visible attached to the toy truck. Question: “Is there a rope in this image?” Model answer: *No*. Correct audit: the failure is **over-withholding**, the rope is small but visibly present; “Cannot tell” is appropriate only if the rope cannot be resolved at full zoom.

Part 1, Object-Presence Audit (per item: image + object)

- Q1. Is a {object} visibly present in this image?
☐ Yes ☐ No ☐ Cannot tell

- Q2. How confident are you?
☐ Certain ☐ Fairly confident ☐ Guessing
- Q3. (If No) Is there a *similar or related* object that could plausibly be mistaken for it (e.g., a bench for a chair)?
☐ Yes, namely: _____ ☐ No

Part 2, Error-Direction Attribution (per item: image + question + model answer + gold label)

- Q4. The model’s answer disagrees with the gold label. Which description best fits the failure?
☐ **Over-claiming**: the model asserted something the image does not show
☐ **Over-withholding**: the model denied something the image does show
☐ **Gold label appears wrong**: the image contradicts the gold label
☐ Cannot tell
- Q5. (Optional, free text) Briefly, what in the image most likely misled the model? _____

Part 3, Caption Audit (per item: image + one-sentence caption)

- Q6. Does the caption mention any object or entity that is **not** visible in the image?
☐ Yes, namely: _____ ☐ No ☐ Cannot tell
- Q7. Does the caption **omit** any prominent object a one-sentence description should reasonably include?
☐ Yes, namely: _____ ☐ No
- Q8. Rate the caption’s fluency (grammar and naturalness), ignoring factual accuracy.
☐ 5 Excellent ☐ 4 ☐ 3 ☐ 2 ☐ 1 Unreadable
- Q9. Rate the caption’s overall faithfulness to the image.
☐ 5 Fully faithful ☐ 4 ☐ 3 ☐ 2 ☐ 1 Mostly wrong

Part 4, Debrief (once per session)

- Q10. Did any items feel ambiguous in a way the options did not capture? (free text)
- Q11. Approximately how many items did you skip or guess on? (number)

I Implementation and Environment Details

Table 20: Hyperparameters (shared across all ablation arms; referenced from Section 5.1).

LoRA rank r	16	LoRA α / dropout	32 / 0.05
Optimizer / lr	AdamW / 10^{-4} , 3×10^{-5}	NPO β	0.1
Steps (fixed / loop)	150 / ≤ 300	Probe period K	25 / 20
Controller gain γ	0.6 / 0.3	Dead-band ε	0.1
$p_{\mathcal{U}}$ bounds	[0.15, 0.85]	Probe / eval size	80 / 250
Image long side	512 px	Seed	0

Software. PyTorch 2.10 (CUDA 12.8), Transformers 5.5.0, PEFT 0.19.1, Unsloth 2026.7.4 (fast Qwen2-VL / Qwen2.5-VL / LLaVA patching), bfloat16 throughout. **Hardware.** NVIDIA H100 80GB;

one GPU per pipeline; PYTORCH_CUDA_ALLOC_CONF=expandable_segments=True

Model configuration. LoRA on language-tower attention projections (q_proj, k_proj, v_proj, o_proj); vision tower and MLP untouched; images resized to a 512-px long side before the processor. This resizing is not a confound: re-evaluating the base model at native resolution changes full-split adversarial accuracy by under 1pp (61.7% to 62.7%) and leaves the signed score unchanged (+0.992), so all reported conclusions hold at native resolution. The NPO reference policy is realized by disabling the LoRA adapter in-place, no second model copy, no extra memory. **Wall-clock.** Diagnosis over 8k queries: 1.5–2.5 h per model on a shared GPU (25 min unshared); each 150-step ablation including two 250-query evaluations: under 1 h (2B model); closed-loop runs bounded by 300 steps plus 12 probe evaluations.

J Artifact Manifest and Reproduction Commands

All artifacts are available at the anonymized repository (<https://anonymous.4open.science/r/DUAL-Compass-DD12>); the pilot’s full pipeline is four scripts and one JSON output per run.

Table 21: Artifact manifest.

Script / artifact	Role
gen_extra_queries.py	rule-based expansion (App. C)
expand_any.py	diagnosis pass $\rightarrow$ streams (Sec. 4.1)
train_dual_unsloth.py	fixed-weight DUAL + ablation arms
train_dual_loop.py	compass-guided closed loop (Sec. 4.3)
eval_utility.py	caption utility audit (App. F)
dual_full_{tag}.jsonl	per-model stream file
result_{tag}_{mode}.json	per-run metrics (before/after/delta)
adapter_{tag}_loop/	trained LoRA adapter

Reproduction of Table 1 (after the diagnosis pass):

```
python train_dual_unsloth.py -mode {dual|unlearn_only|learn_only} \
    -steps 150 -max_per_stream 120 -eval_n 250
```

All randomness is controlled by seed 0 (Python, PyTorch); evaluation is sampling-free.

K Extended Qualitative Examples

Table 22 gives diagnosed records across all four sources. Over-claims are consistently co-occurrence-plausible (a cowboy hat in a ranch scene, a kitchen island in a kitchen, medals at a ceremony); misses include both fine-grained objects (a trackpad, a CD case) and scene-level ones (trees, a dining table).

The extended sample sharpens both regularities. Over-claims are uniformly *scene-schema completions*: a life jacket on a boat, a traffic light at an intersection, a winch on machinery, coffee beans near a coffee product, the model completes a scene schema rather than verifying visual evidence, the same co-occurrence mechanism DASH mines at web scale [4], here identified per instance and tagged with its direction. Misses skew toward *background and support surfaces* (floor, table, sky) and *partially occluded mid-ground objects* (truck, backpack), suggesting the no-bias stream teaches attention to unremarked regions rather than rare categories. That

Table 22: Extended diagnosed records (Qwen2-VL-2B, full pool).

Stream	Source	Object	Model error
unlearn	NoCaps	cowboy hat	asserted; absent
unlearn	NoCaps	kitchen island	asserted; absent
unlearn	NoCaps	medals	asserted; absent
unlearn	AMBER	acacia tree	asserted; absent
unlearn	AMBER	saddle	asserted; absent
unlearn	VizWiz	snack bag	asserted; absent
learn	NoCaps	net	denied; present
learn	AMBER	drink	denied; present
learn	AMBER	trees	denied; present
learn	VizWiz	trackpad	denied; present
learn	VizWiz	CD case	denied; present
learn	COCO	dining table	denied; present
unlearn	AMBER	life jacket	asserted; absent
unlearn	COCO	traffic light	asserted; absent
unlearn	NoCaps	winch	asserted; absent
unlearn	AMBER	passenger train	asserted; absent
unlearn	VizWiz	coffee beans	asserted; absent
unlearn	AMBER	beach ball	asserted; absent
learn	VizWiz	table	denied; present
learn	NoCaps	floor	denied; present
learn	AMBER	sky	denied; present
learn	VizWiz	floor (2nd img.)	denied; present
learn	COCO	truck	denied; present
learn	COCO	backpack	denied; present

the two streams have such different visual anatomies is itself an argument for treating them with different losses.

L Probe-Sensitivity Analysis

Table 23 contrasts the two measurement regimes on the same base model and pool. Greedy decoding reports a moderate over-claim problem; the log-likelihood probe reveals that on adversarial absent objects the model’s latent preference favors assertion in 80.9% of cases, for many queries the two answers are nearly equiprobable, and the sampled “no” conceals a marginal preference. The two regimes agree on the *direction* (both $s > 0$) but differ five-fold in magnitude. Since re-ranking, option scoring, and temperature sampling all consult the underlying likelihoods, we treat the latent probe as the deployment-relevant stress measure and greedy decoding as the user-visible one; both are reported wherever feasible.

Table 23: Greedy vs. latent (log-likelihood) measurement, Qwen2-VL-2B base model.

Regime / pool	FP rate	FN rate	s
Greedy, full 4,006-query pool	14.3%	6.5%	+0.386
Latent, 250 held-out adversarial	80.9%	0.7%	+0.978

M Gradient Analysis of the DUAL Objective

This appendix analyzes the three terms of Eq. (2) at the gradient level, explaining (i) why NPO-based unlearning is stable but weak

in isolation, (ii) why the learn term in isolation collapses the model, and (iii) qualitatively how the controller of Eq. (3) balances the two.

M.1 Adaptive Damping of the Unlearn Term

Write $r = \log \frac{\pi_\theta(y^-|v,x)}{\pi_{\text{ref}}(y^-|v,x)}$ for the policy-to-reference log-ratio of the hallucinated assertion. The NPO term is $\ell_u(r) = -\frac{2}{\beta} \log \sigma(-\beta r)$, with

$$\frac{\partial \ell_u}{\partial r} = 2 \sigma(\beta r) \in (0, 2). \quad (5)$$

The gradient magnitude on the suppressed assertion is gated by $\sigma(\beta r)$: as unlearning progresses and $\pi_\theta(y^-)$ falls below the reference ($r \rightarrow -\infty$), the gate closes exponentially. This is the collapse-slowing property inherited from NPO [39], and it also explains the near-null result of the unlearn-only arm in Table 1: from the very first step the term only ever *reduces* the assertion probability toward (and below) the reference, it contains no force that raises the probability of the *correct* answer, so a model whose assertion prior is strongly skewed retains its ranking between “Yes.” and “No.” almost everywhere (s : +0.978 to +0.977). Suppression alone therefore cannot rebalance the assertion prior.

M.2 Why the Learn Term Alone Collapses

The learn stream contains only records whose gold answer is Yes. (missed present objects). In isolation, its cross-entropy gradient $-\nabla_\theta \log \pi_\theta(\text{“Yes.”} | v, x)$ raises the probability of the answer-initial token *Yes* for every training context. Because the LoRA update acts on shared attention projections, not on per-image circuits, a portion of this update is context-independent: it moves the language-tower’s answer prior toward assertion in *all* presence-question contexts, including those where the correct answer is No. With no counter-pressure in the batch (no absent-object records at all), the context-independent component accumulates unopposed, and the model reaches the degenerate optimum of the stream, answering “yes” unconditionally (FP 100%, Table 1). The retain stream alone cannot prevent this, because retain records are dominated by the model’s existing correct behavior and carry little gradient once fitted. The failure is thus structural, a direction-unbalanced batch is a biased objective, which is the gradient-level restatement of the paper’s central claim.

M.3 The Controller as Proportional Balance Feedback

Let $\delta_u > 0$ denote the (average) reduction in the model’s assertion margin caused by one unlearn step, and $\delta_l > 0$ the increase caused by one learn step, both measured on the probe distribution. Between consecutive probes, the expected drift of the signed score satisfies, to first order,

$$\mathbb{E}[\Delta s_t] \propto -(p_{\mathcal{U}}(t) \delta_u - q_{\mathcal{L}}(t) \delta_l), \quad (6)$$

where $q_{\mathcal{L}}$ is the learn-draw probability. The proportional rule $p_{\mathcal{U}} = \text{clip}(0.5 + \gamma s_t, \cdot)$ makes the drift sign-opposing: yes-biased states ($s_t > 0$) increase unlearn pressure and vice versa, so $s = 0$ is the unique attractor within the clip region, and the dead-band $|s_t| \leq \varepsilon$ converts the attractor into a stopping set. Because δ_u and δ_l are unknown and drift as the adapter trains, we make no quantitative convergence claim; the fixed-weight run of Table 1 ($p_{\mathcal{U}}$ constant,

no stop) empirically demonstrates the failure mode the feedback removes, the state passes through $s = 0$ and continues to -0.657 .

N Notation

Table 24: Notation used throughout the paper.

Symbol	Meaning
$\pi_\theta, \pi_{\text{ref}}$	subject policy; frozen reference (adapter off)
v, x	image; text query
y^+, y^-	gold answer; hallucinated assertion (“Yes.”)
$\mathcal{U}, \mathcal{L}, \mathcal{R}$	unlearn / learn / retain streams
FP, FN	over-claims; misses
s, s_t	behavioral signed score (Eq. 1); at probe t
λ_u, λ_r	unlearn / retain loss weights
β	NPO inverse-temperature
$p_{\mathcal{U}}(t)$	unlearn-draw probability set by Eq. (3)
K, γ, ϵ	probe period; controller gain; dead-band
$\mathcal{P}$	held-out balanced probe set

O Extended Benchmark Comparison

Table 25 situates the evaluation resources discussed in Section 2 along task format, primary target, and whether failure *direction* is an explicit output of the benchmark’s protocol. Direction-awareness in the sense used by this paper, a signed quantity that can route an intervention, appears only in the suite we build on; BEAF’s change-based metrics (stubbornness, indecision) are the nearest neighbors, characterizing answer instability under scene edits rather than a signed assertion bias.

Table 25: Evaluation resources for VLM hallucination. “Dir.” = does the protocol output an explicit failure-direction signal?

Resource	Venue	Format	Primary target	Dir.
CHAIR [24]	EMNLP’18	caption	object recall	–
POPE [16]	EMNLP’23	yes/no	object presence	Δ
NOPE [19]	ALVR’24	VQA	negative presence	–
AMBER [28]	arXiv’23	mixed	attr./relation	–
HallusionBench [8]	CVPR’24	VQA	illusion/language	–
MMHal-Bench [27]	ACL’24(F)	free-form	response quality	–
BEAF [33]	ECCV’24	yes/no $\times 2$	change consistency	Δ
HALLUCINOGEN [25]	UncNLP’25	implicit	yes-bias escalation	Δ
DASH [4]	ICCV’25	retrieval	systematic FPs	–
HalluCompass [2]	u.r.’26	bin.+free	signed direction	✓

P Behavioral Streams and the Judge-Side Adjective Partition

The training signal in this paper is behavioral (FP/FN membership). The judge-based diagnosis of the underlying suite instead extracts failure-describing *adjectives* from a judge LLM and maps them through a frozen 16-word partition. Table 26 reproduces the partition and aligns it with our streams: yes-bias adjectives describe exactly the behavior our unlearn stream suppresses, and no-bias adjectives the behavior our learn stream repairs; generic adjectives

correspond to perception failures that both streams leave untouched (they surface in either stream depending on the direction the misperception happened to take). This alignment is what lets either diagnosis serve as the compass (Section 6): the streams identify *which* examples to treat, the adjectives characterize *what kind* of failure each represents, and the signed scores of the two diagnoses agree at scale (Table 13).

Table 26: The frozen adjective partition of the underlying suite, aligned with DUAL-Compass streams.

Class	Adjectives	Stream
YesBIAS	overconfident, overinclusive, optimistic, careless, misinformed, biased	$\mathcal{U}$ (unlearn)
NoBIAS	overcautious, partial, incomplete, superficial, overlooked	$\mathcal{L}$ (learn)
GENERIC	misperceptive, misrecognized, misinterpreting, misleading, inaccurate	either

Q Dataset Licenses and Compute Budget

Licenses. All image bytes derive from the underlying suite’s release and retain their original licenses: MS-COCO val2014 (CC-BY 4.0), AMBER (research-only redistribution), NoCaps validation (CC-BY-SA), VizWiz validation (CC-BY 4.0). Derived queries, stream files, adapters, and code will be released under CC-BY 4.0 (data) and MIT (code), matching the upstream suite. No new images, personal data, or human-subject data are collected; VizWiz images (taken by blind users) are used strictly under their released research terms.

Compute budget. Table 27 itemizes the GPU cost of every experiment in this paper. The full pilot, diagnosis, expansion, three ablation arms, closed loop, and utility audit for the primary model, fits within approximately twelve H100-hours; the cross-model replication approximately doubles this. For comparison, preference-based alignment pipelines report an order of magnitude more for the feedback stage alone [30].

Table 27: Approximate compute budget (H100, shared-node wall-clock; unshared times in parentheses where measured).

Phase	GPU-hours
Diagnosis pass, 4k queries (2B)	1.7
Diagnosis pass, 8k queries, per model $\times 3$	1.5–2.5 each
Ablation arm (150 steps + 2 evals) $\times 3$	<1 each
Closed-loop run (≤ 300 steps + 12 probes), per model	1–2 each
Utility audit (80 captions), per model	0.3 each
Primary-model pilot total	≈ 12

R Case-Study Walkthrough

We trace one real record end-to-end to make the pipeline concrete.

1. Query. Image AMBER_267.jpg (an equestrian scene); query “Is there a saddle in this image? Answer with only ‘yes’ or ‘no.’”; gold

label *no* (the annotation’s `hallu_adv` list contains *saddle*: plausible near a horse, verified absent).

2. Diagnosis. Greedy decoding yields “Yes”. The record enters the unlearn stream $\mathcal{U}$ with direction tag `yes_bias`, an auditable statement: *the model over-claimed a contextually plausible absent object*.

3. Training step. When drawn (probability $p_{\mathcal{U}}(t)$), the NPO term computes the summed log-likelihood of Yes. under the adapter and under the adapter-disabled reference; the gated gradient ($2\sigma(\beta r)$; Appendix M) lowers the assertion’s likelihood ratio. No gradient ever targets No. on this image, suppression, not substitution, so related true assertions (*horse*, *fence*) are untouched except through shared parameters, which the retain stream anchors.

4. Post-training audit. After training, the record is re-scored. Under the DUAL arm of Table 1, records of this type flip to *No* in the large majority of held-out cases (FP 80.9% to 5.5%); the per-direction repair counts reported by the training script state exactly how many $\mathcal{U}$ -type and $\mathcal{L}$ -type failures were repaired, persisted, or newly introduced.

Contrast. The same record under the learn-only arm receives no suppression at all; moreover, the global assertion drift induced by the unbalanced learn stream (Appendix M) makes this over-claim *more* likely, the mechanism behind the 100% FP collapse.

S Threats to Validity

Table 28: Limitations and the evidence that bounds each (referenced from Section 6).

Limitation	Status / mitigating evidence
Object-presence scope	Free-form generation measured via utility audit (§5.5) and human study; span-level extension is future work
External transfer (scope)	Claims are on-distribution; cross-distribution transfer to DASH-B hard negatives is unsolved by in-suite or DASH-train supervision (App. U) — an explicit boundary, not a claim
Judge dependence	Training uses behavioral diagnosis (no judge); adjective judge agrees in sign at scale (Table 13)
Seed sensitivity	Main arms reported over 3 seeds (Table 1: unlearn 61.2±3.8, learn 56.0±0.0, DUAL fixed 82.4±4.6); the closed loop removes the fixed-weight seed variance; some appendix ablations remain single-seed
Probe distribution	Suite-drawn probe can under-represent adversarial bias (2B); documented and quantified (§5.9)
Image resolution	512-px training; native-resolution re-evaluation shifts accuracy <1pp (App. I)

Internal. All ablation arms share hyperparameters, data pool, and step budget, so Table 1 differences are attributable to the objective; the main arms are reported over three seeds (the ordering is seed-invariant), though some appendix ablations remain single-seed and step-budget interactions (e.g., whether unlearn-only would move at 10× steps) are unexplored. The learn-only collapse is an extreme point: gentler learn-dominant mixtures would collapse more

slowly, and we have not mapped the full mixture–outcome surface. **Construct.** The behavioral signed score treats all over-claims and misses as exchangeable within their direction; adjective-level diagnosis (Appendix P) distinguishes failure kinds within a direction, and per-kind repair analysis is future work. The automatic caption metric is substring-based and conservative (Appendix F); the human audit (Section 5.8) bounds its error and independently confirms the quality gain (89% of captions judged hallucination-free, 85% preferred over the baselines). **External.** Evidence covers 2–7B open VLMs on object presence over four public image sources; extrapolation to frontier-scale models, attribute/relational hallucination, and non-English queries is unsupported by our data. Streams and evaluation come from one suite’s disjoint splits; cross-benchmark transfer (DASH-B, BEAF) is planned. **Statistical.** Held-out evaluation uses 250 queries; the headline effects (75.4pp FP reduction, 44pp collapse gap between arms) are far outside binomial noise at this n (± 6 pp at 95%), but small deltas (e.g., unlearn-only’s +0.8pp) are within it and we do not interpret them.

T Full Results Table

Table 29 consolidates every training run reported in the paper on the common held-out adversarial protocol, for audibility. Fixed-weight rows vary only in data order/seed and step budget; their spread is the fragility the controller removes.

Table 29: All 2B training runs (held-out adversarial). s = signed score after training; base is +0.98 / 64.0%.

Method	Config	Acc.	s
Unlearn only (NPO)	150 st.	64.8%	+0.98
Learn only (SFT)	150 st.	56.0%	+1.00
Balanced SFT	seed 0	62.8%	−0.96
Balanced SFT	seed 1	44.8%	−1.00
Balanced SFT	seed 2	73.2%	−0.70
Fixed DUAL (NPO)	order A	86.0%	−0.66
Fixed DUAL (NPO)	seed 1	77.2%	−0.90
Fixed DUAL (NPO)	seed 2	84.0%	−0.55
Fixed DUAL (NPO)	450 st.	71.6%	−0.75
Fixed GA (0.3)	450 st.	74.0%	−0.97
Fixed GA (1.0)	150 st.	44.0%	−1.00
Fixed NPO ($\lambda_u=3$)	150 st.	44.4%	−1.00
Loop + ckpt. (NPO)	damped	86.4%	+0.53
Loop + ckpt. (GA 0.3)	damped	86.8%	+0.35
Loop + ckpt. (GA 1.0)	damped	86.8%	+0.21
Loop, undamped (NPO)	$\gamma=0.6$	44.0%	−1.00

U Cross-Distribution External Evaluation

U.1 DASH-B: an honest cross-distribution negative

On DASH-B [4] (web-mined systematic false positives, held-out 1,000-example split), no intervention we tried yields a genuine, non-collapsed improvement in external true-negative rate from in-suite or DASH-train supervision: a fixed-weight DASH-heavy run reaches 100% TNR only by collapsing to universal denial (TPR

8.9%), and a distribution-matched controller balances the DASH log-likelihood probe yet still greedy-decodes yes-biased (TNR 27.2%), the latent-vs-greedy gap of Appendix L now across distributions. We report this as an honest limitation that bounds the method's scope to the diagnosed distribution; the constructive counterpart, where diagnosing *on* the deployment distribution does repair it, is the COCO result below.

U.2 Distribution-Matched (COCO) Repair Protocol

This appendix documents the constructive experiment of Section 5.9 (Table 6). The official POPE-COCO pool of 500 images is split deterministically (sorted by filename) into the first 300 for diagnosis and training and the disjoint last 200 for held-out evaluation, *no training image appears at evaluation*. The subject model is diagnosed on the 300 training images' presence queries (greedy), producing direction streams; because both Qwen models are no-biased on COCO the streams are learn-heavy (2B: 11 unlearn / 1,287 learn / 4,102 retain; 3B: 174 / 498 / 4,728). The compass loop then trains with a COCO-drawn balanced probe and best-balance checkpointing (same hyperparameters as the main loop, Table 20). Evaluation is greedy on the held-out 200 images (400 queries per POPE split). Base signed scores on this 200-image split (−0.97 for 2B, −0.46 for 3B) are milder than the full-500 scores in Table 7 (−0.99, −0.62) because the subset is smaller and differently sampled.

V Reproducibility Checklist

- **Models.** All subject models are public: Qwen2-VL-2B-Instruct, Qwen2.5-VL-3B-Instruct, llava-1.5-7b-hf (Unsloth mirrors).
- **Data.** All images, annotations, and queries are from the public HalluCompass release; DASH-B from its public release. The direction-balanced expansion is rule-based (App. C) and deterministic under seed 0.
- **Code.** Five scripts (App. J) reproduce diagnosis, expansion, fixed/loop training, and all evaluations; released under MIT.
- **Determinism.** Evaluation is sampling-free (log-likelihood comparison). Training uses seed 0 unless a seed is stated; we note the unsloth model-load reseeding pitfall and re-seed immediately before stream shuffling.
- **Hardware / cost.** Single H100 per run; full primary-model pilot ≈ 12 GPU-hours (App. Q).
- **Hyperparameters.** Complete in Table 20; identical across ablation arms.
- **Metrics.** FP/FN rate, accuracy, signed score (Eq. 1) on held-out adversarial; standard greedy POPE-compatible (acc/P/R/F1/yes-ratio); DASH-B TNR/TPR; caption hallucination/length/degeneration for utility.
- **Statistics.** Multi-seed for decisive comparisons; headline effects far exceed the ± 2.7 pp binomial band at the full-split $n=1,341$.
- **Known limitations.** Enumerated in Table 28 and App. S.

W Broader Impact

DUAL-Compass targets a deployment-safety failure (fabricated visual assertions) while explicitly protecting the complementary utility failure (missed objects), and it makes the intervention auditable at the level of failure direction. Positive impact: teams can repair a diagnosed bias with a low-cost, transparent procedure rather than an opaque uniform pressure, and can verify *which* direction they moved. Risks and mitigations: (i) any behavioral fine-tune can shift unrelated capabilities, we recommend re-running full safety evaluations post-training and provide a utility audit as a first screen; (ii) a controller optimized to a mis-specified probe distribution can balance the wrong thing, we document the probe-distribution effect and recommend diagnosing on the deployment distribution; (iii) the method reduces but does not eliminate hallucination, and should not be represented as a safety guarantee. No new models, scraped data, or personal data are released; all artifacts derive from public research datasets.